

Tooling. Python 3 / pandas / numpy / scipy / matplotlib; all estimates computed programmatically, none typed by hand; analysis code, figure code and write-up drafted with Claude Code.

How to read the charts. Each kind of panel answers one question.

Chart	Question it answers
Histogram / PDF	What is the shape: where is the peak, how long is the tail?
PMF (airports)	How many airports have exactly k routes? (Route counts are whole numbers.)
CDF	What share lies at or below x ? Where is the largest gap D between data and model?
CCDF	What share lies at or above x ? On log-log axes a power law is a straight line.
QQ plot	Does each quantile of the model match the data's? Points on $y = x$ mean yes.
Box plot (movies)	Are the centre, spread and range the same? Are there impossible ratings outside 1 to 10?

Linear axes show where most values are; log-log axes spread out values that span several orders of magnitude.

Problem 1.

Airport data:

(a) What is the α if the distribution is power law?

$\alpha = 1.612$, with $x_{\min} = 1$ route. Maximum-likelihood estimate $\hat{\alpha} = 1 + n / \sum_i \ln(x_i / x_{\min})$, $x_{\min} = \min_i x_i$, $n = 3409$.

$n = 3409$	number of airports
$x_{\min} = \min_i x_i = 1$	smallest route count
$S = \sum_{i=1}^{3409} \ln(x_i / 1) = 5569.43$	sum of log ratios
$n / S = 3409 / 5569.43 = 0.6121$	
$\hat{\alpha} = 1 + 0.6121 = 1.612$	estimate

(b) What is the λ if the distribution is exponential?

$\lambda = 0.0504$. MLE $\hat{\lambda} = 1 / \bar{x} = 1 / 19.845$.

$\sum_i x_i = 67652$	total routes
$\bar{x} = 67652 / 3409 = 19.845$	mean
$\hat{\lambda} = 1 / 19.845 = 0.0504$	estimate

(c) What is $[a, b]$ if the distribution is uniform?

$[a, b] = [1, 915]$, the minimum and maximum route counts (MLE).

$\hat{a} = \min_i x_i = 1$	fewest routes
$\hat{b} = \max_i x_i = 915$	most routes

(d) What is μ and σ if the distribution is normal?

$\mu = 19.845$, $\sigma = 53.506$ (sample mean and sample standard deviation with $n - 1$).

$$\hat{\mu} = \bar{x} = 19.845$$

mean, as in the exponential fit

$$\sum_i (x_i - \bar{x})^2 = 9,756,906.22$$

sum of squared deviations

$$9,756,906.22 / (3409 - 1) = 2862.94$$

sample variance

$$\hat{\sigma} = \sqrt{2862.94} = 53.506$$

standard deviation

(These can be CDF, PDF, or QQplot)

Note on the axes. In every chart on linear axes, the routes-per-airport axis is cut at 200 on purpose: 98% of airports have at most 200 routes, and on a full 0–915 axis they would be squeezed into the left edge. The whole distribution, including the tail up to 915 routes and the model samples beyond it, is shown in the log-log chart next to each linear one. The share of the data and of each model sample beyond 200 is printed on the linear panels.

For (e) to (i):

- **Samples.** $n = 3409$ random values drawn from each fitted model with numpy's random generator (seed 20701313; one generator for all eight samples, so every figure can be rebuilt exactly).
- Power law: numpy has no power-law sampler, so I used inverse transform sampling. Draw u uniformly from $[0, 1)$ and set $x = x_{\min}(1 - u)^{-1/(\alpha-1)}$, which solves $u = F(x)$ for the fitted CDF $F(x) = 1 - (x/x_{\min})^{-(\alpha-1)}$.
- Exponential: `rng.exponential(1/λ)` (numpy takes the mean $1/\lambda$, not λ).
- Uniform: `rng.uniform(a, b)`. Normal: `rng.normal(μ, σ)`.

The samples are real numbers, not whole route counts: they contain values such as 1.08, and the normal sample contains negative values.

- **Colours.** Data in blue, model sample in orange.
- **Layout of the airport figures.** Rows PDF, CDF, CCDF and QQ, with linear axes on the left and log-log axes on the right; the data's median and mean are marked in every distribution panel, and each CDF/CCDF shows its value at $x = 1$.
- **D.** Two-sample Kolmogorov–Smirnov distance: the largest vertical gap between the two CDFs (0 = identical).

(e) Screen shot of the visualization of distribution of data

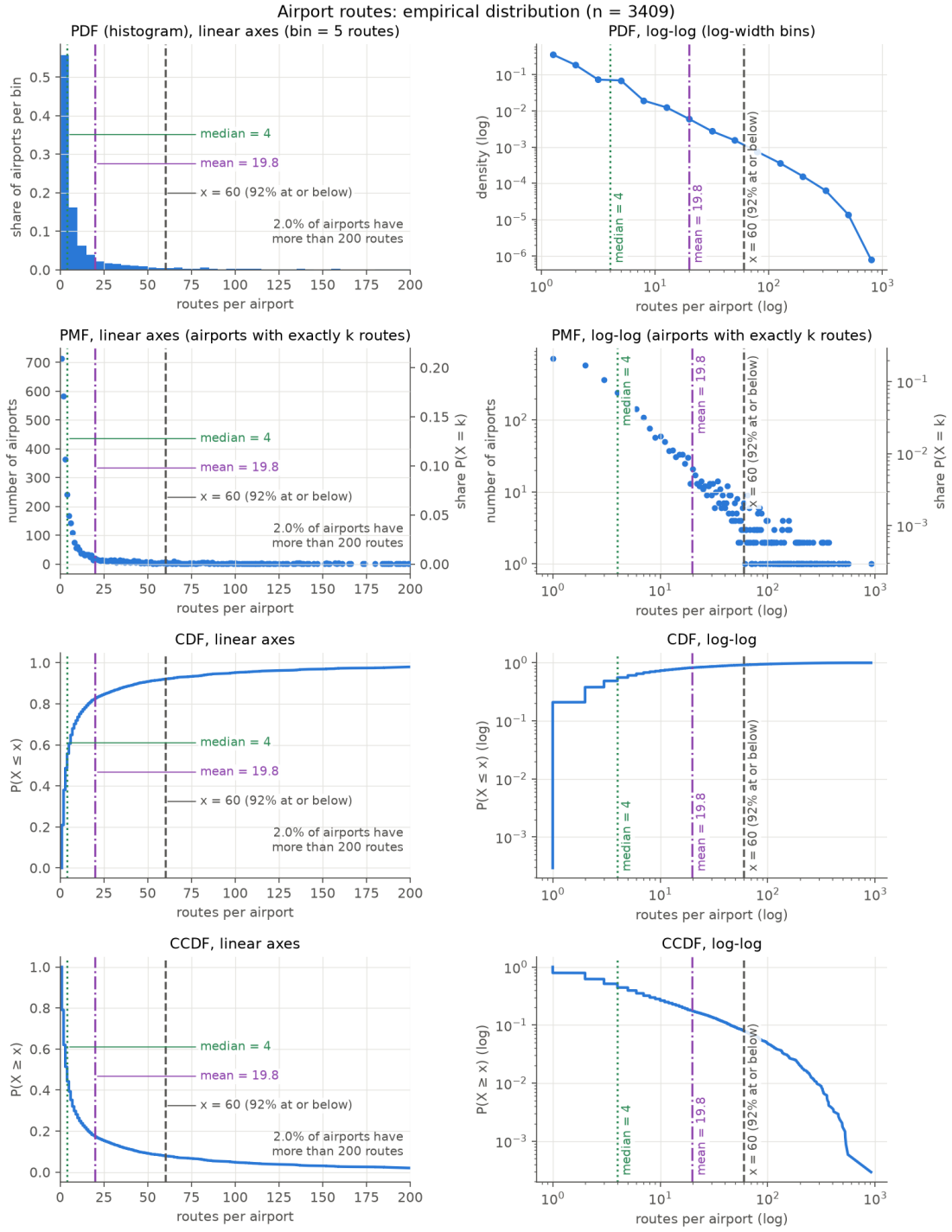


Figure 1: Airport routes: empirical distribution ($n = 3409$). Rows: PDF, PMF, CDF, CCDF. Left column on linear axes, right column on log-log axes.

(f) Screen shot of the visualization of distribution of data if it was power law

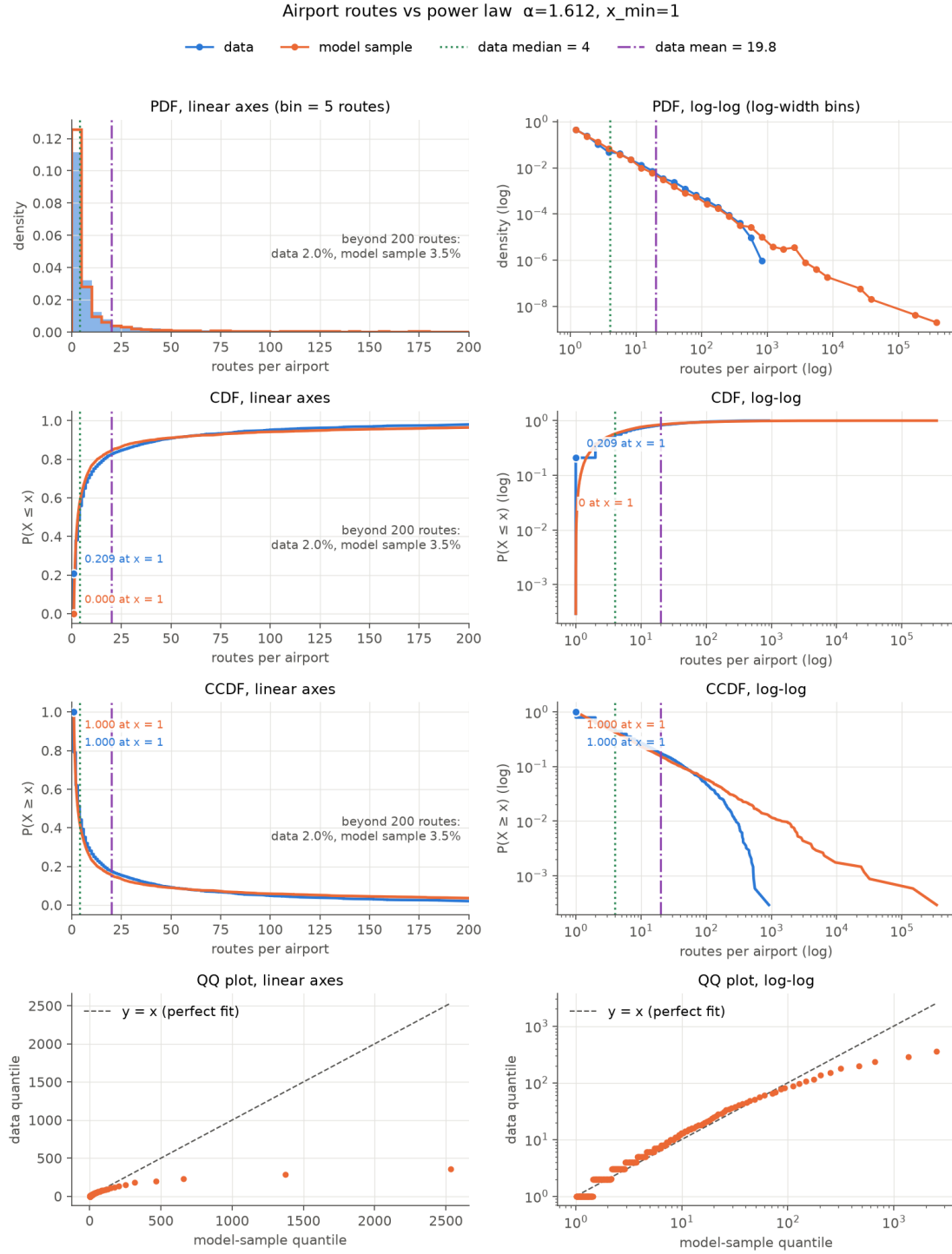


Figure 2: Airport routes vs. power law ($\alpha = 1.612$, $x_{\min} = 1$). $D = 0.209$.

(g) Screen shot of the visualization of distribution of data if it was exponential

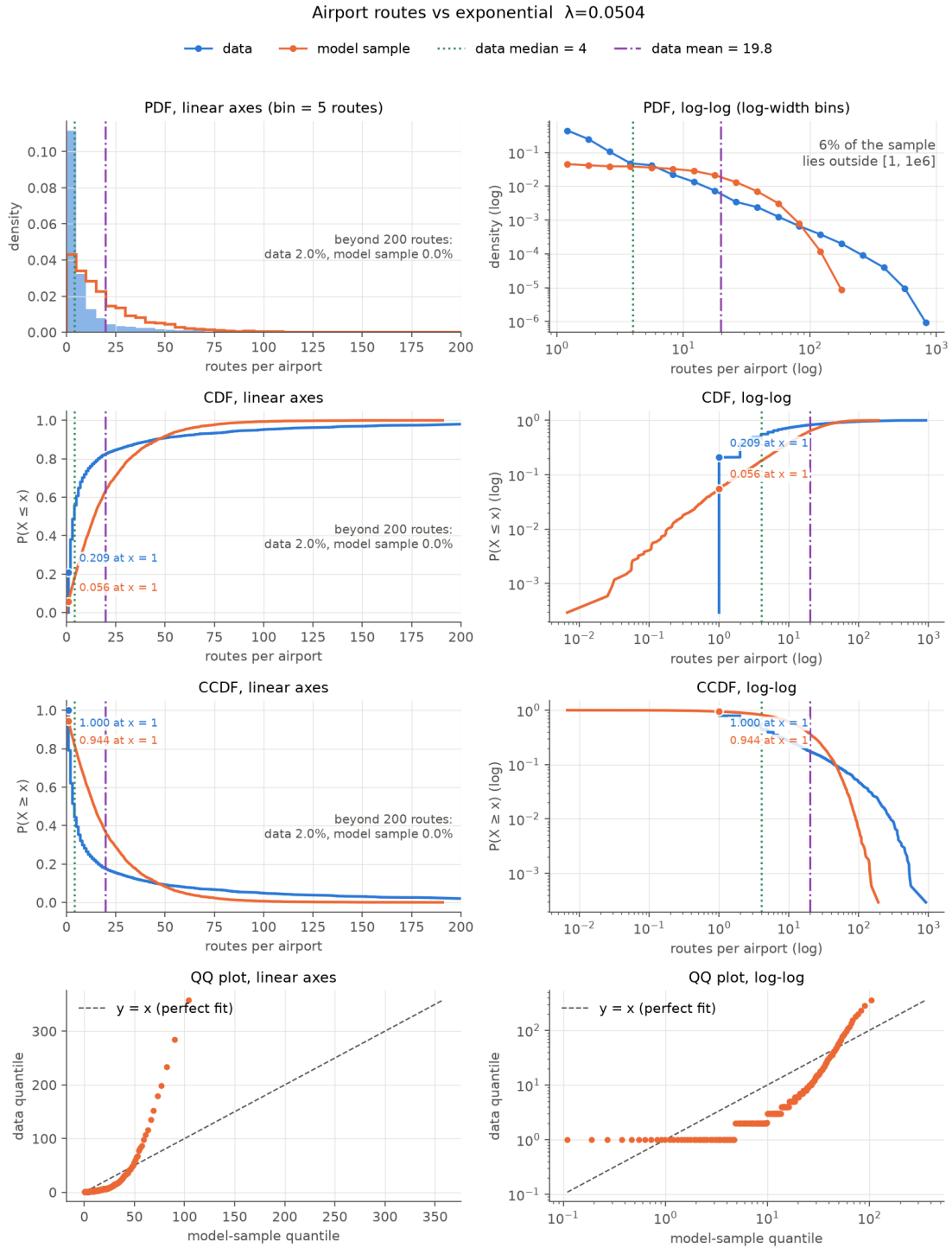


Figure 3: Airport routes vs. exponential ($\lambda = 0.0504$). $D = 0.397$.

(h) Screen shot of the visualization of distribution of data if it was uniform

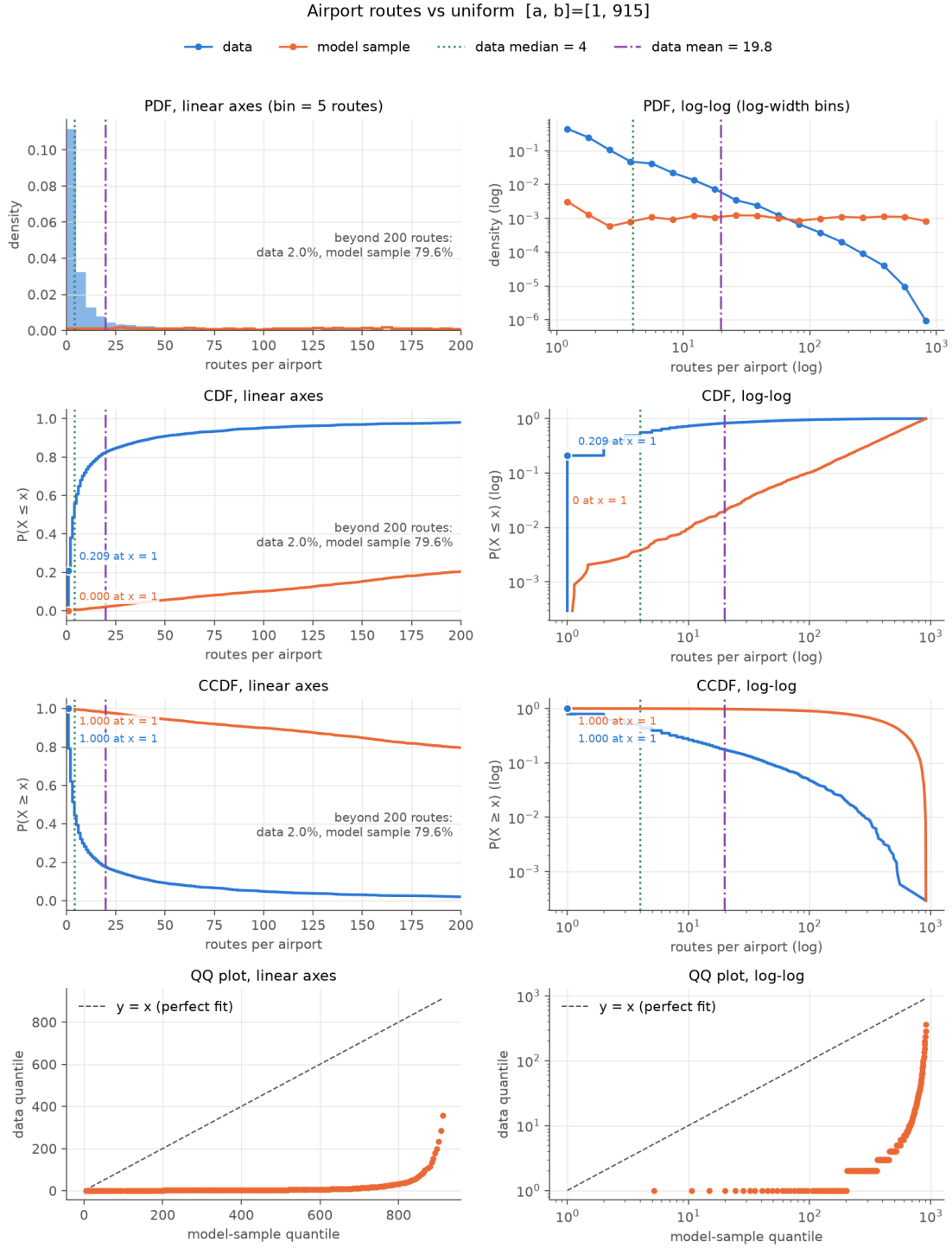


Figure 4: Airport routes vs. uniform on [1, 915]. $D = 0.856$.

(i) Screen shot of the visualization of distribution of data if it was normal

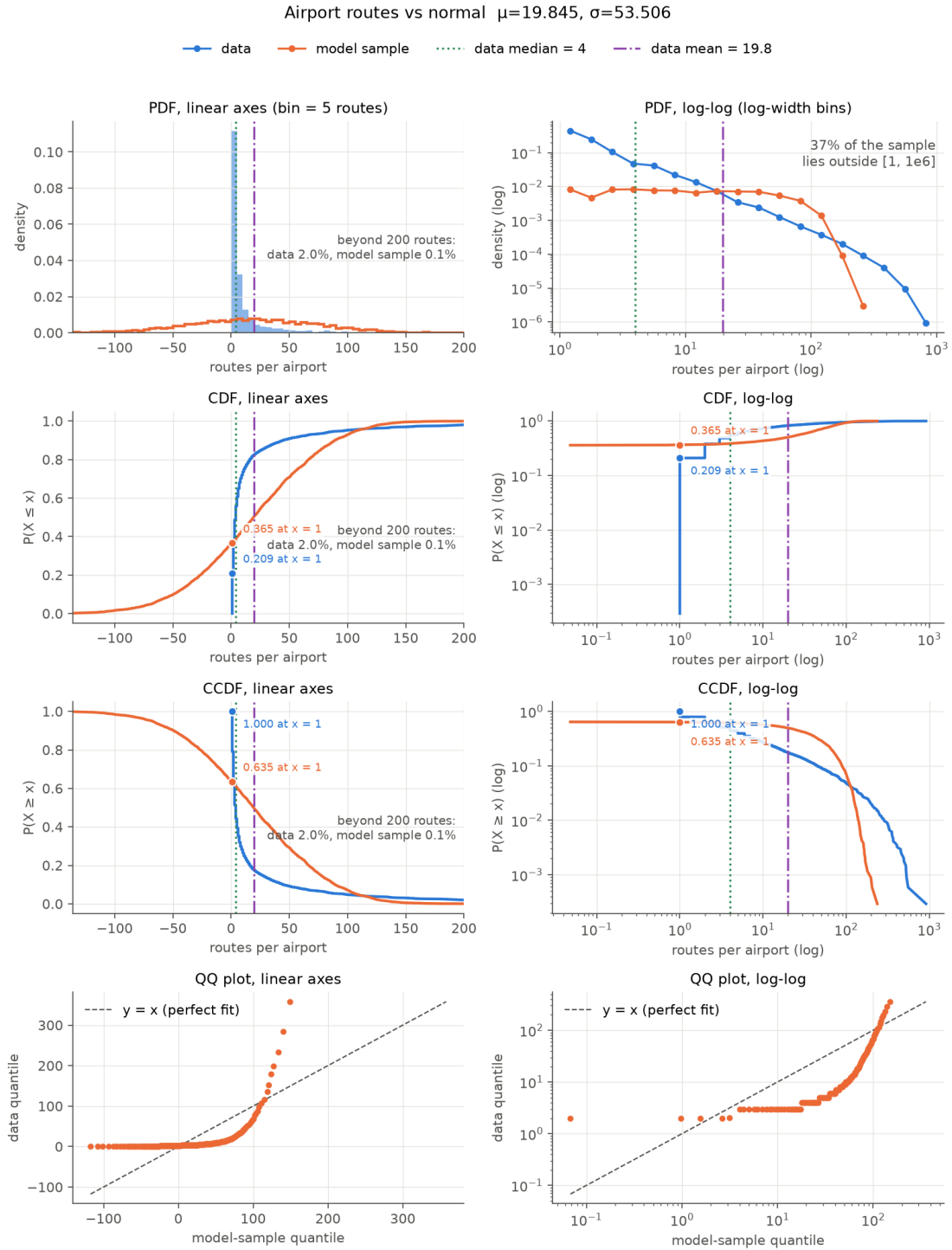


Figure 5: Airport routes vs. normal ($\mu = 19.8, \sigma = 53.5$). $D = 0.365$.

(j) What is the distribution of data based on your observation? Discuss

	power law	exponential	uniform	normal
KS distance D (data vs model sample)	0.209	0.397	0.856	0.365
where the gap is largest, x	1	6	60	0.98
data CDF $F_{\text{data}}(x)$ there	0.209	0.648	0.922	0.000
model-sample CDF $F_{\text{model}}(x)$ there	0.000	0.251	0.065	0.365

How D is computed. D is the two-sample Kolmogorov–Smirnov distance from `scipy.stats.ks_2samp`, computed on the raw values: the largest vertical gap between the data’s CDF and the model sample’s CDF (0 = identical, 1 = no overlap),

$$D = \max_x |F_{\text{data}}(x) - F_{\text{model}}(x)|.$$

Conclusion: the airport data follows a power law.

Why the power law fits (Figure 2):

- **Lowest D :** 0.209, against 0.365 to 0.856 for the other three models.
- **PDF shape:** the data and the model sample match on both kinds of axes. On linear axes the bars have almost the same heights; on log-log axes both follow the same straight line up to about 300 routes.
- **QQ plot (log-log):** the points lie on $y = x$ up to about 60 routes.
- **Skewness:** the data is strongly right-skewed (6.0), and only the power law is that skewed. Over the data’s range (up to 915 routes) its sample has 6.8; the exponential’s has 2.0, the uniform’s and the normal’s 0. (With its giant sampled airports included the power-law sample reaches 47.5: for $\hat{\alpha} < 4$ the model’s skewness is infinite.)
- **80/20:** the top 20% of airports hold 83.8% of all routes, close to the 80/20 rule that power laws are known for. The power-law sample gives 87.1% over the data’s range; the exponential gives 51.9%, the normal 41.9% (positive values only) and the uniform 35.6%. This is the same fact as the skewness seen from another side: a few hubs with hundreds of routes carry most of the total, which is also why the mean (19.8) is five times the median (4).

Why the others fail: their QQ plot shape is different; the points leave $y = x$ almost from the start.

- **Exponential** (Figure 3, $D = 0.397$): too steep, below the line for small airports and above it for hubs.
- **Uniform** (Figure 4, $D = 0.856$): every point far below the line.
- **Normal** (Figure 5, $D = 0.365$): too steep, with negative model quantiles that no airport has.

Movie Rating data:

(a) What is the α if the distribution is power law?

$\alpha = 1.851$, with $x_{\min} = 1.9$ (the lowest average vote). Same MLE as for the airports, $n = 4392$.

$n = 4392$	number of movies
$x_{\min} = \min_i x_i = 1.9$	lowest average vote
$S = \sum_{i=1}^{4392} \ln(x_i/1.9) = 5163.93$	sum of log ratios
$n/S = 4392/5163.93 = 0.8505$	
$\hat{\alpha} = 1 + 0.8505 = 1.851$	estimate

(b) What is the λ if the distribution is exponential?

$\lambda = 0.1606$. MLE $\hat{\lambda} = 1/\bar{x} = 1/6.227$.

$\sum_i x_i = 27348.7$	sum of votes
$\bar{x} = 27348.7/4392 = 6.227$	mean
$\hat{\lambda} = 1/6.227 = 0.1606$	estimate

(c) What is $[a, b]$ if the distribution is uniform?

$[a, b] = [1.9, 8.5]$, the minimum and maximum votes (MLE).

$\hat{a} = \min_i x_i = 1.9$	lowest vote
$\hat{b} = \max_i x_i = 8.5$	highest vote

(d) What is μ and σ if the distribution is normal?

$\mu = 6.227$, $\sigma = 0.893$ (sample mean and sample standard deviation with $n - 1$).

$\hat{\mu} = \bar{x} = 6.227$	mean, as in the exponential fit
$\sum_i (x_i - \bar{x})^2 = 3503.28$	sum of squared deviations
$3503.28/(4392 - 1) = 0.7978$	sample variance
$\hat{\sigma} = \sqrt{0.7978} = 0.893$	standard deviation

(These can be CDF, PDF, or QQplot)

Note on the axes.

- All movie charts use linear axes; average votes lie between 1.9 and 8.5.
- In (f) to (i) the vote axis is cut at 20 on purpose, so the data's peak stays visible. The power-law and exponential samples go far beyond any possible rating (the power law's 99.5th percentile is about 900).
- Each panel prints the share of the model sample beyond the axis.
- For those two models the QQ plot uses log axes, so the whole sample is shown.

For (f) to (i):

- **Samples.** $n = 4392$ random values drawn from each fitted model in the same way as for the airports (same generator, continuing after the airport samples). The samples are not rounded to 0.1 like the votes.
- **Colours.** Data in blue, model sample in orange.
- **Layout of the movie figures.** PDF and CDF on top, QQ plot and box plot below. The CDF marks where the largest gap D occurs; the box plot shows median, quartiles and whiskers, with dashed lines at the possible ratings 1 and 10.
- D . Two-sample Kolmogorov–Smirnov distance: the largest vertical gap between the two CDFs (0 = identical).

(e) Screen shot of the visualization of distribution of data

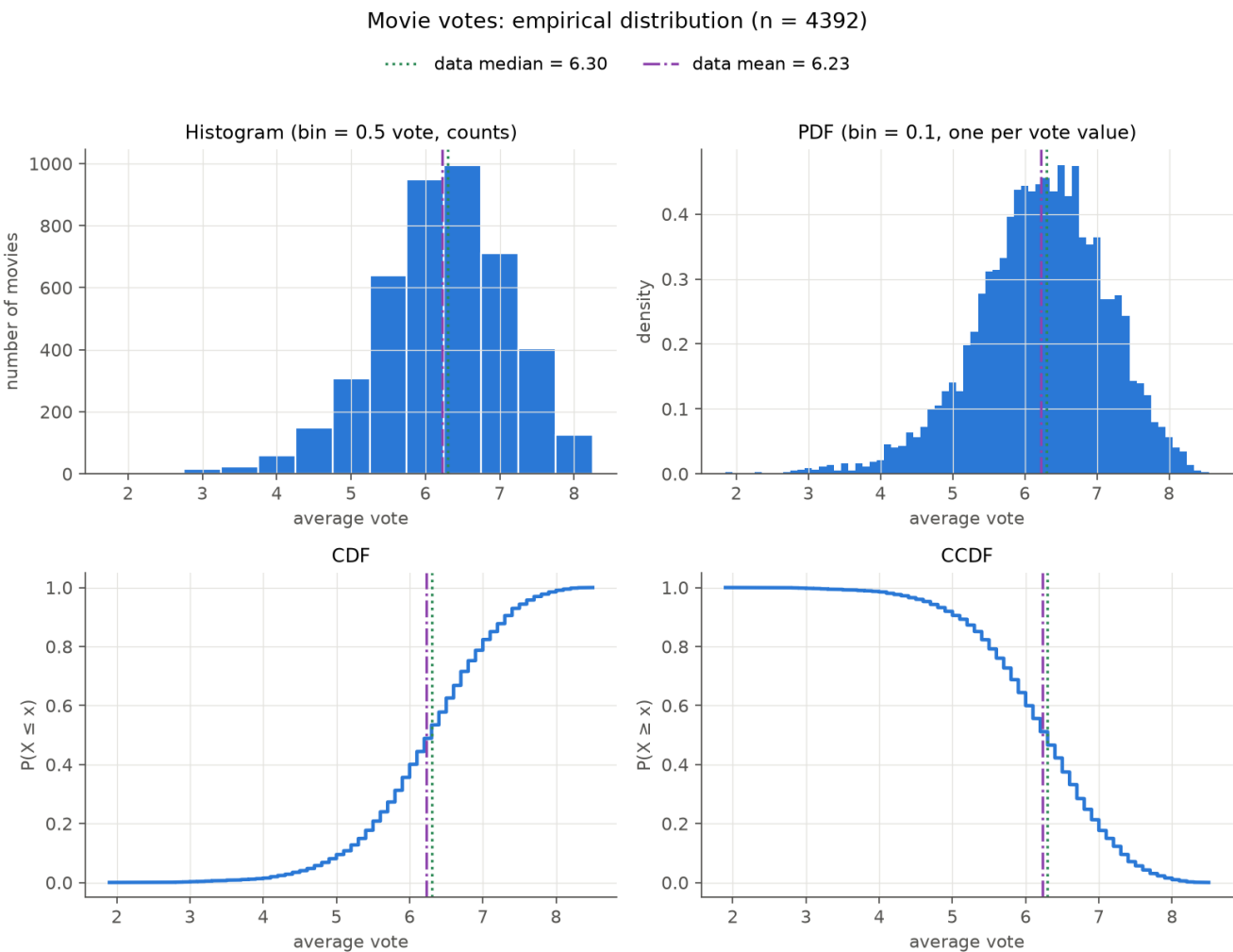


Figure 6: Movie votes: empirical distribution ($n = 4392$). Histogram (counts), PDF, CDF and CCDF, all on linear axes.

(f) Screen shot of the visualization of distribution of data if it was power law

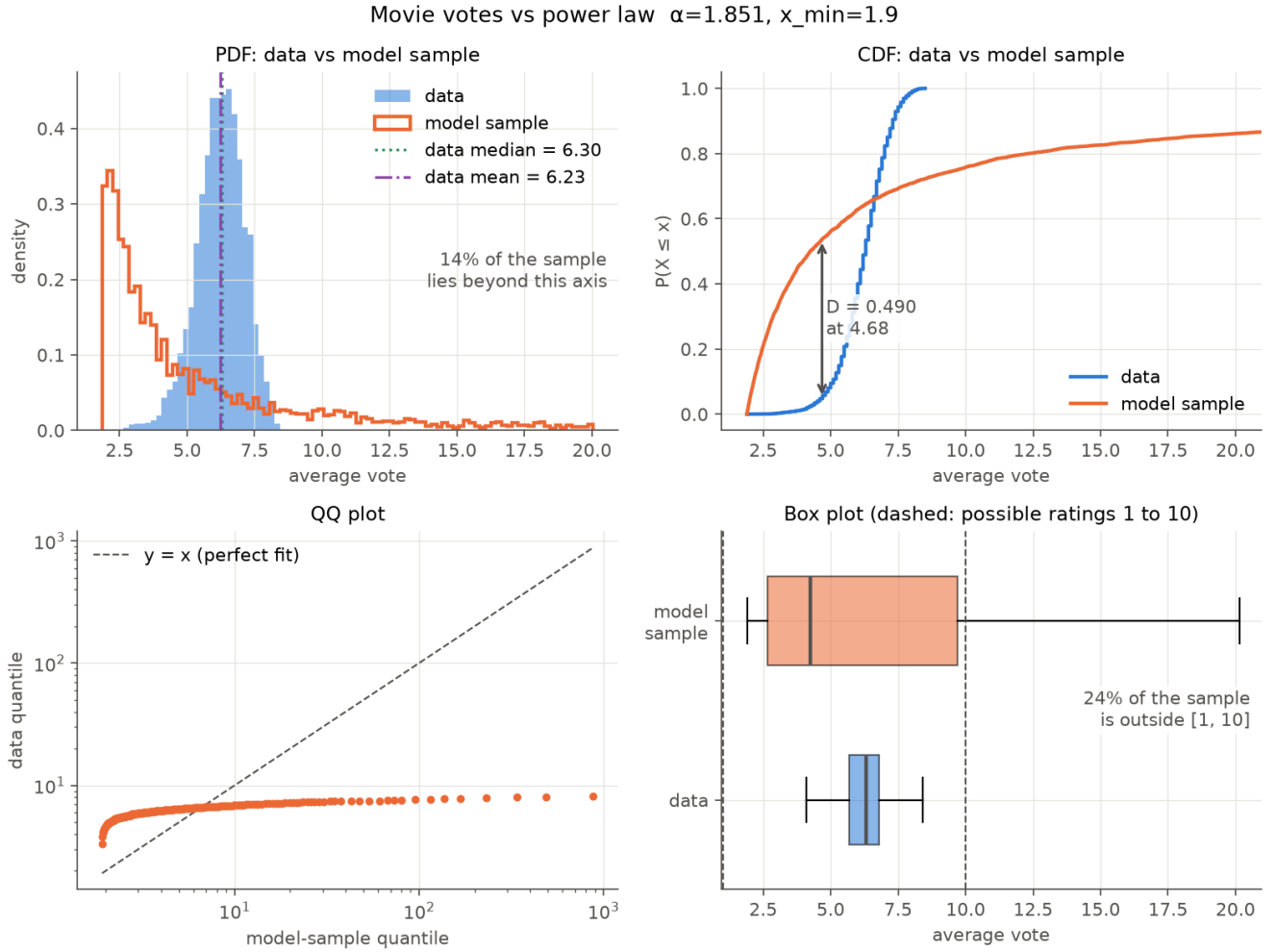


Figure 7: Movie votes vs. power law ($\alpha = 1.851$, $x_{\min} = 1.9$). $D = 0.490$.

(g) Screen shot of the visualization of distribution of data if it was exponential

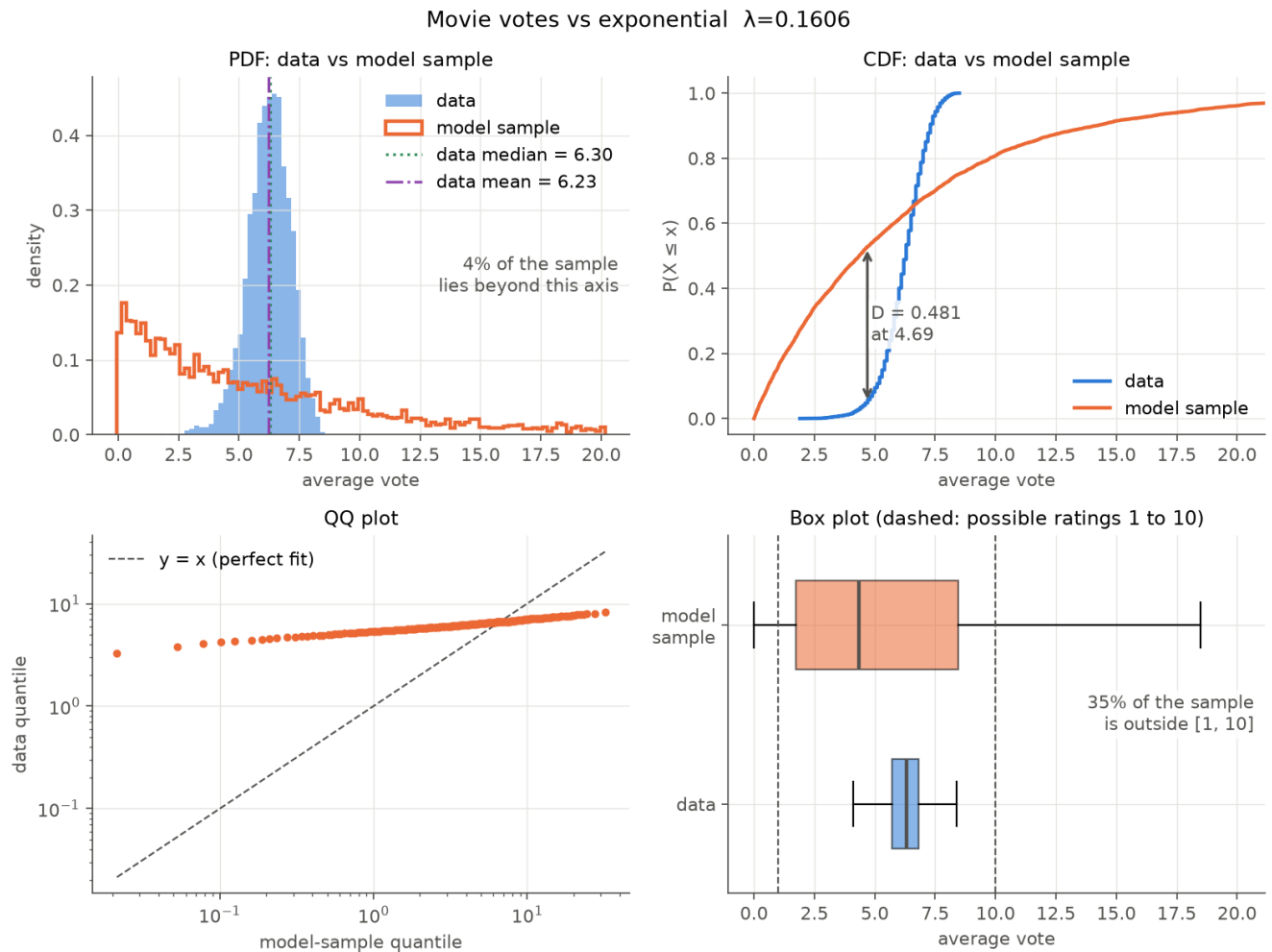


Figure 8: Movie votes vs. exponential ($\lambda = 0.1606$). $D = 0.481$.

(h) Screen shot of the visualization of distribution of data if it was uniform

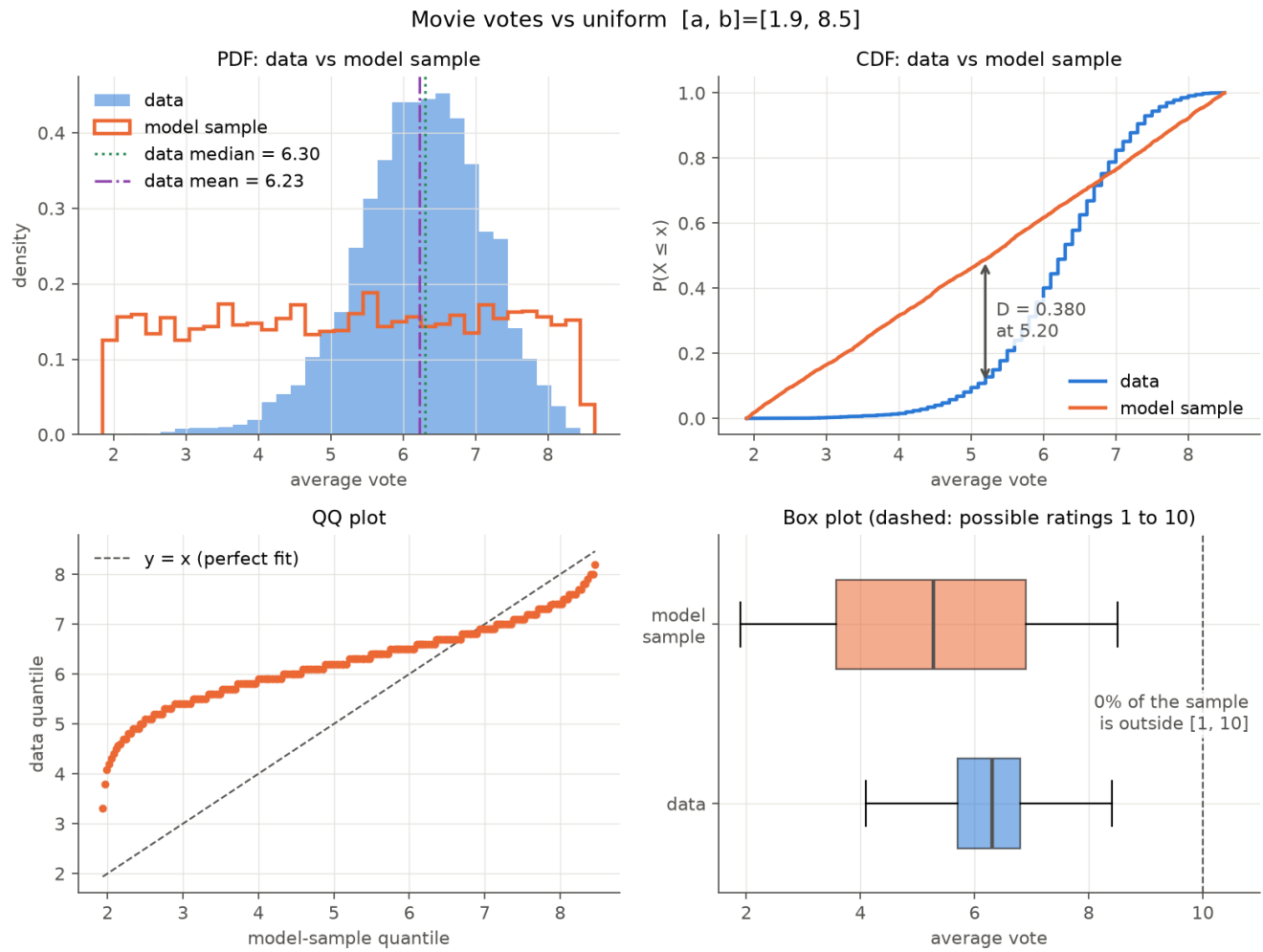


Figure 9: Movie votes vs. uniform on [1.9, 8.5]. $D = 0.380$.

(i) Screen shot of the visualization of distribution of data if it was normal

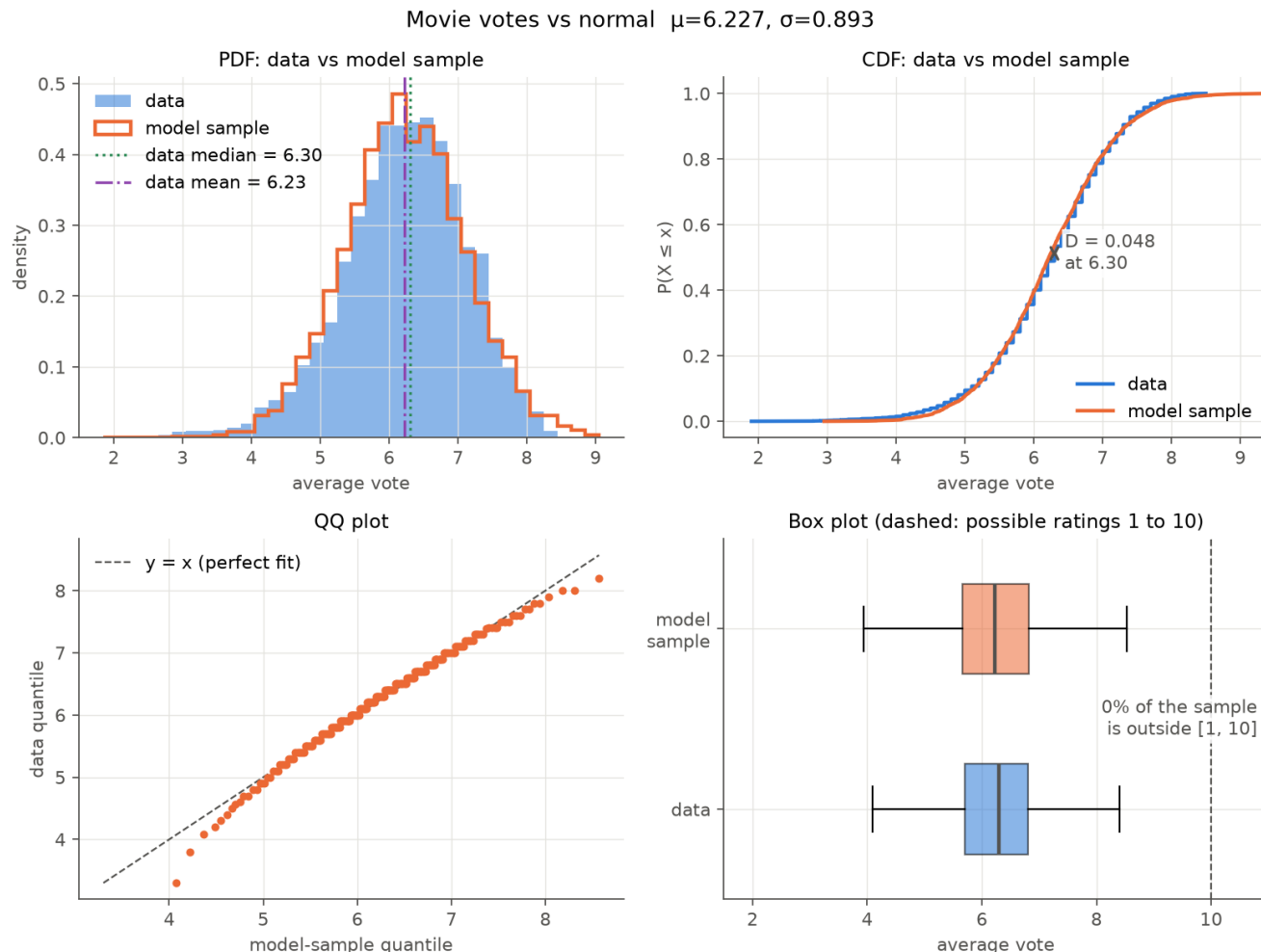


Figure 10: Movie votes vs. normal ($\mu = 6.227$, $\sigma = 0.893$). $D = 0.048$.

(j) What is the distribution of data based on your observation? Discuss

	power law	exponential	uniform	normal
KS distance D (data vs model sample)	0.490	0.481	0.380	0.048
where the gap is largest, x	4.7	4.7	5.2	6.3
data CDF $F_{\text{data}}(x)$ there	0.047	0.047	0.107	0.489
model-sample CDF $F_{\text{model}}(x)$ there	0.538	0.528	0.488	0.536

How D is computed. D is the two-sample Kolmogorov–Smirnov distance from `scipy.stats.ks_2samp`, computed on the raw values: the largest vertical gap between the data’s CDF and the model sample’s CDF (0 = identical, 1 = no overlap),

$$D = \max_x |F_{\text{data}}(x) - F_{\text{model}}(x)|.$$

Conclusion: the movie votes follow a normal distribution, approximately.

Why the normal fits (Figure 10):

- **Smallest gap:** $D = 0.048$, about one eighth of the next-best model (0.380).
- **Same shape:** the sample traces the data's single bell, with the peak in the same place near 6.3.
- **QQ on the line:** the points lie on $y = x$ from about 4.5 to 7.5.
- **Same spread:** 70.1% of movies are within $\pm 1\sigma$ and 95.6% within $\pm 2\sigma$; the normal predicts 68.3% and 95.4%. The box plots nearly coincide.
- **Expected:** each vote is an average of many ratings, and by the central limit theorem such averages tend toward a normal distribution.

Why the others fail: their PDF shape is totally different from the data's single bell.

- **Power law** (Figure 7, $D = 0.490$): falls from the lowest vote.
- **Exponential** (Figure 8, $D = 0.481$): falls from 0.
- **Uniform** (Figure 9, $D = 0.380$): flat, no peak.

Problem 2.

- Screenshot of distribution/histogram of the age

How the ages were computed.

- **Date format:** dd-Mon-yy. 277 rows use a 4-digit year instead: 129 are the placeholder 0000 (dropped, see below) and 148 are real years from 1732 to 1899, which are kept as written. All 148 give ages of 127 or more, so the age filter removes them later.
- **Century:** every 2-digit year is read as 19yy. IMDB-WIKI was collected in 2015, and the counts per year fall smoothly down to 00 with no jump into the 2000s.
- **Missing dates:** 129 rows have year 0000 (the year is unknown), so no age can be computed and they are dropped.
- **Age:** 2026 minus birth year, month and day ignored (professor's instruction, 2026-09-23).
- **Filter:** keep $1 \leq \text{age} \leq 99$, as the assignment asks. 7,500 rows are over 99 and dropped; none are under 1.
- **Result:** 453,089 people; mean 56.79, median 55, range 27 to 99. The minimum is 27 because the latest birth year is 1999 ($2026 - 1999 = 27$).

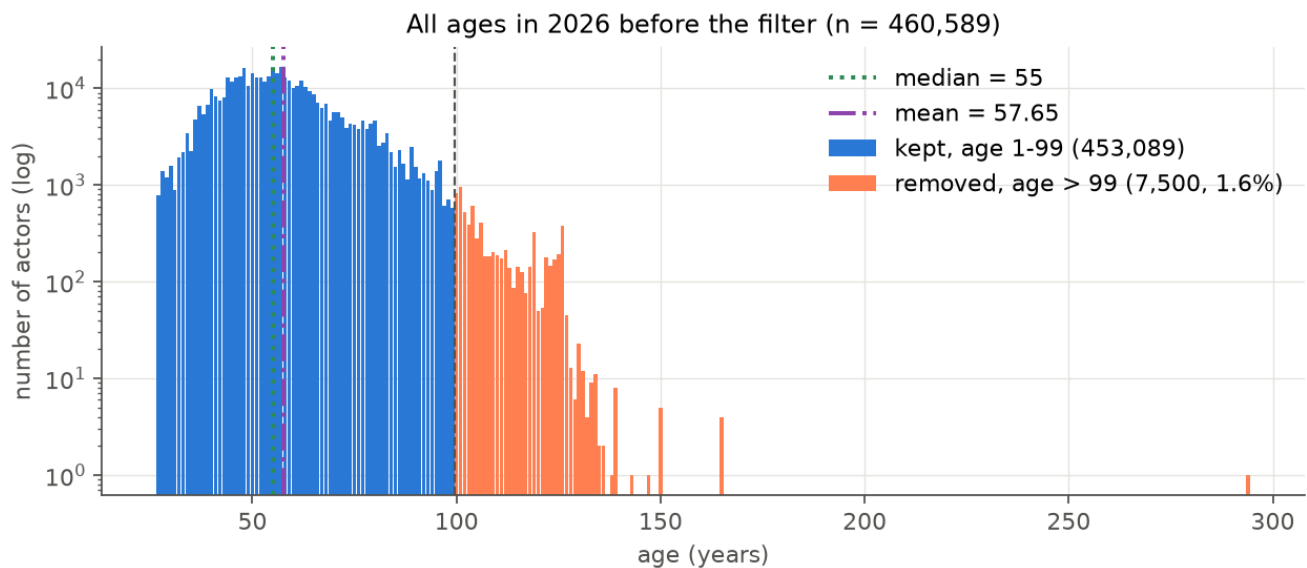


Figure 11: All ages in 2026 before the $1 \leq \text{age} \leq 99$ filter ($n = 460,589$). Blue bars are kept, coral bars are removed; log y axis. Median 55 (green dotted), mean 57.65 (purple dash-dot).

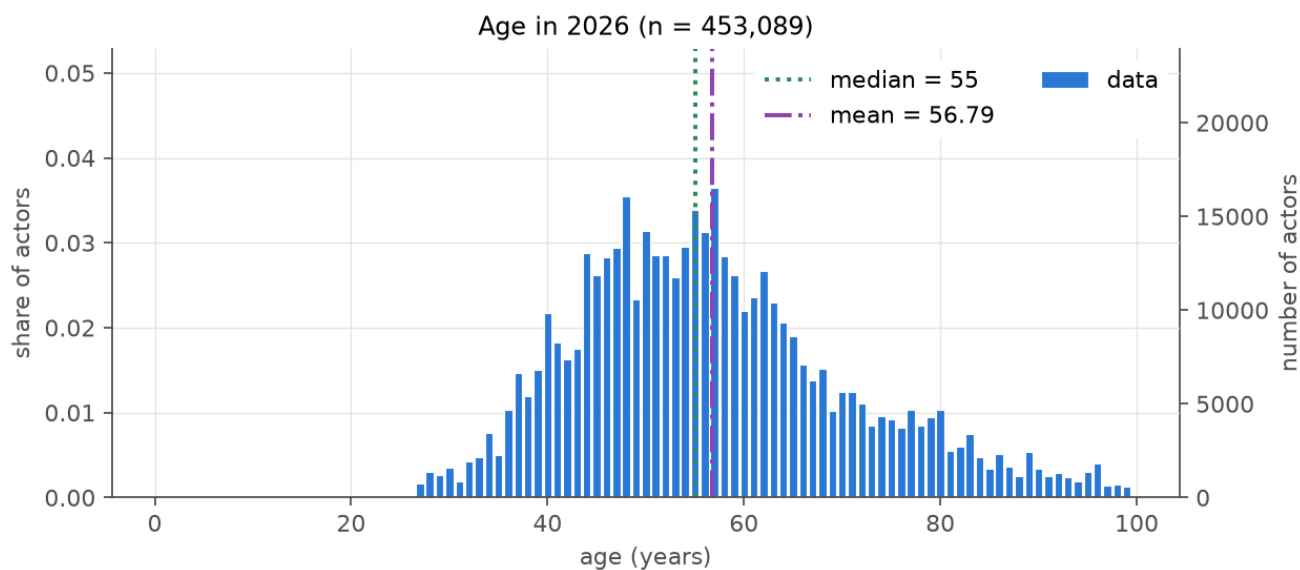


Figure 12: Age in 2026 ($n = 453,089$). Median 55 (green dotted), mean 56.79 (purple dash-dot).

- Screenshot of distribution/histogram of the first digit of age

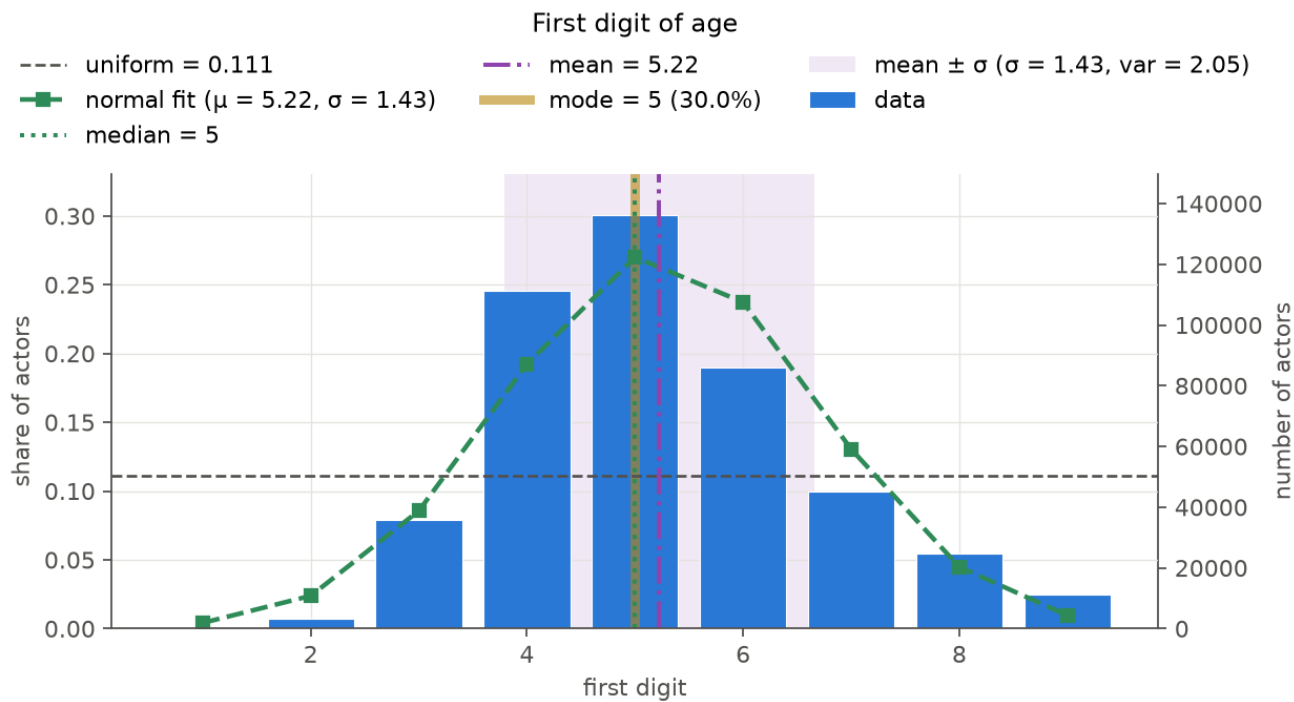


Figure 13: First digit of age, with the uniform level $1/9$ (grey dashed) and a normal fitted to the digits (green dashed, $\mu = 5.22, \sigma = 1.43$). Median 5 (green dotted), mean 5.22 (purple dash-dot), mode 5 (gold); the shaded band is mean $\pm \sigma$ ($\sigma = 1.43$, variance 2.05).

- Screenshot of distribution/histogram of the last digit of age

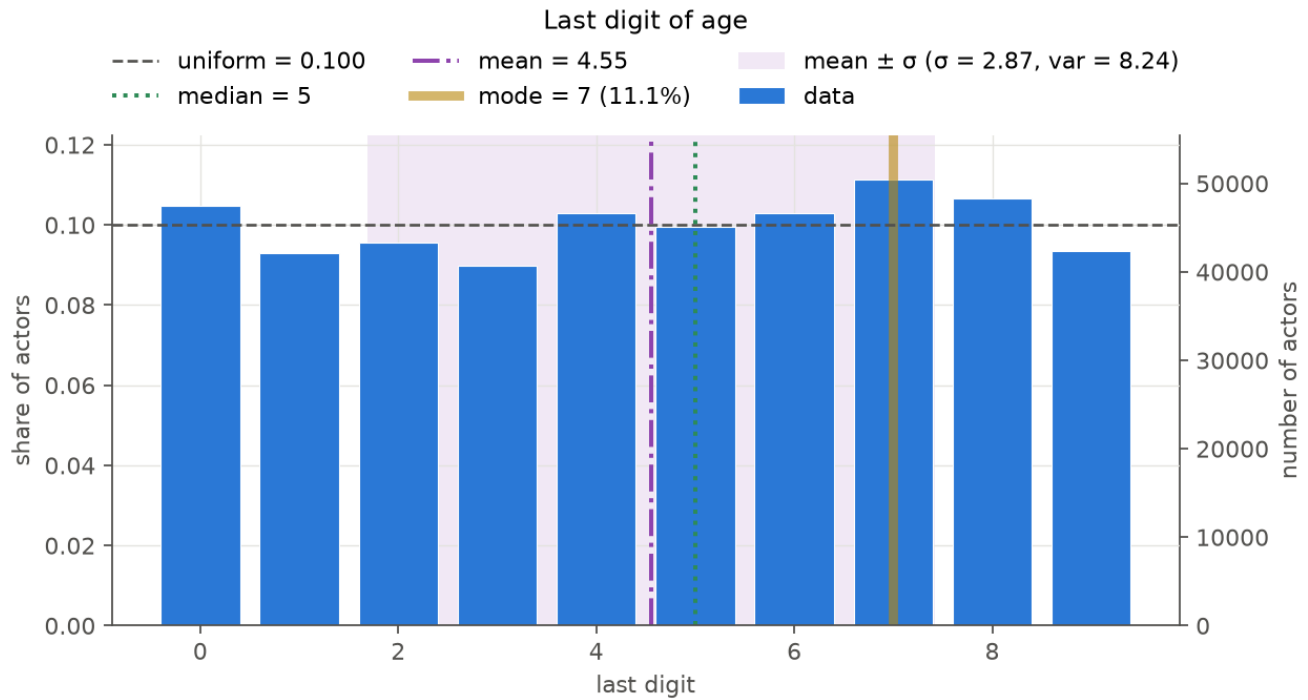


Figure 14: Last digit of age, with the uniform level 1/10 (dashed). Median 5 (green dotted), mean 4.55 (purple dash-dot), mode 7 (gold); the shaded band is mean $\pm \sigma$ ($\sigma = 2.87$, variance 8.24).

- Discussion (what are those distribution?) Why do you think it is like this?

The first digit is approximately normal; the last digit is approximately uniform.

First digit: approximately normal, because:

1. It is bell-shaped, with one peak at 5 that falls off on both sides.
2. The mean (5.22), median (5) and mode (5) are all in the middle and almost equal.
3. A normal fitted to the digits ($\mu = 5.22$, $\sigma = 1.43$) follows the bars within 5.3 points (Figure 13).
4. For ages 30 to 99 the first digit is the decade of age, and age itself has one hump (Figure 12).
5. It is not exactly normal: it is slightly right-skewed (+0.53), digit 2 is small because ages are measured in 2026, so the youngest actor (born 1999) is already 27, and digit 1 never appears because no age is between 10 and 19.

Last digit: approximately uniform, because:

1. The bars are flat, with every digit within $10\% \pm 1.2$ points (Figure 14).
2. The mean (4.55) and variance (8.24) match a uniform on 0–9 (4.5 and $(10^2 - 1)/12 = 8.25$).
3. The last digit is the age modulo 10, so every block of ten ages contains each digit once, and age counts change slowly within a block.
4. The small gaps are year-to-year noise in births (Figure 12), not a preference for any digit.